

Siddhanta Adhikari

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🐙 GitHub 🔗 LinkedIn

PROFILE

Final-year Computer Engineering student with a strong foundation in core CS principles and hands-on experience building end-to-end systems. Experienced in developing Machine Learning and full-stack applications to solve complex technical challenges, ranging from predictive modeling to secure authentication. Hackathon winner with a focus on translating technical requirements into functional, production-ready solutions.

EDUCATION

Bachelor of Engineering in Computer Engineering
Kathmandu University

2026
Dhulikhel, Nepal

SKILLS

Programming Languages

Python, C++, JavaScript, C#

Frameworks & Libraries

PyTorch, TensorFlow, OpenCV, Django, Qt Framework

Databases

PostgreSQL, MongoDB, SQLite

Core Concepts

Machine Learning, Artificial Intelligence, Deep Learning, Object-Oriented Programming (OOP), Computer Vision, Full-Stack Web Development

Developer Tools

Git, VS Code, Kaggle, Google Colab, Google Earth Engine

CERTIFICATES

- AWS AI Practitioner Challenge [🔗](#)
- CS50's Introduction to Artificial Intelligence with Python [🔗](#)
- Intro to Machine Learning [🔗](#)
- Pandas [🔗](#)

ACADEMIC PROJECTS

Aquila: Satellite-Based Flood Detection

08/2025 – 01/2026

- Spearheaded the development of deep learning models for SAR-based flood detection using Sentinel-1 imagery.
- Benchmarked architectures including UNet, Unet++, ResNet34 and Efficient-net
- Engineered a validation pipeline integrating ESA WorldCover permanent water masks to ensure model reliability.

SFaceX: Facial Recognition & Anti-Spoofing

03/2025 – 07/2025

- Architected a secure authentication module using Dlib and Eye Aspect Ratio (EAR) to detect liveness and prevent spoofing.
- Optimized face detection performance by tuning Haar Cascade classifiers and implementing LBPH for real-time recognition.

StyleMe: Virtual Try-On & Body Measurement

08/2024 – 01/2025

- Engineered a computer vision system using MediaPipe to extract regional body measurements (chest, waist, shoulder) from mobile camera feeds.
- Integrated real-time pose estimation into a Flutter application, allowing for automated clothing size recommendations.
- Resolved high-resolution alignment issues related to human pose occlusions to improve virtual try-on accuracy.

CookeryKaa: Recipe Sharing Platform - Designed comprehensive PostgreSQL database schemas and ER diagrams to support multi-user recipe management. - Built a responsive frontend interface using React and SwiperJS, improving user engagement metrics through interactive UI elements. - Developed secure backend API endpoints using Django for scalable CRUD operations.	02/2024 – 06/2024
Event Ease: Calendar Utility & Converter - Designed the entire frontend architecture using the Qt Framework, ensuring cross-platform responsiveness. - Programmed a complex date conversion algorithm for Bikram Sambat (BS) to AD transitions.	08/2023 – 01/2024

INDEPENDENT PROJECTS

Zero Trust Agent Sandbox <i>Local AI Security Broker & Execution Sandbox • Solo</i> - Engineered a security broker for an AI agent system using Python and Llama 3.2, ensuring every agent action is validated through an independent checkpoint before execution. - Designed an isolated runtime environment using Docker containers with restricted network access and memory limits, eliminating risk of unintended system damage.	06/2026 – 07/2026
Flow <i>Multi-Agent Pathfinding Simulation Core • Group Project</i> - Engineered the primary backend and structural foundation for the autonomous multi-agent simulation. - Built the core grid and spatial mechanics allowing multiple agents to simultaneously initialize and map paths across the simulation field. - Designed the base data structures and state handlers required to track agent properties and movement loops.	05/2026 – 06/2026
Cadence <i>Nepali Folk Music AI & Genre Classifier • Group Project</i> - Collected and processed raw Nepali folk music audio into structured datasets, enabling consistent and reliable model training. - Developed and trained a deep learning classifier combining EfficientNet-B0 and BiLSTM to accurately identify regional folk music genres, benchmarked against an SVM baseline.	04/2026
Aanubadh <i>Document Layout-Preserving Translation Tool • Group Project</i> - Developed the backend document pipeline that converts PDFs to editable formats, applies translations sentence-by-sentence, and rebuilds the layout structurally to avoid text overlapping. - Built the underlying parsing logic to handle raw text streams, ensuring clean segmentation, proper punctuation mapping, and direct integration with low-resource language translation APIs.	05/2026 – 05/2026
MemoryPalace <i>Immersive LLM-Powered Virtual Learning Environment • Group Project</i> - Developed the end-to-end evaluation logic and user testing framework, enabling real-time verification of user knowledge retention within the virtual room. - Engineered system state handlers to track evaluation progress, resolve reassessment bugs, and streamline data flow between frontend inputs and LLM evaluation prompts.	04/2026 – 04/2026

HONORS

Winner: KU Hackathon (AR/VR Track) Conceptualized and built a Unity-based VR application using spatialized audio to guide visually impaired users through 3D environments. Implemented audio-based navigation triggers that respond to user proximity and orientation.	01/11/2025
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RESEARCHS

Spatio-Temporal Flood Prediction MLP-BiLSTM, Google Earth Engine, PyTorch <i>(In Progress)</i> • Developing late fusion MLP-BiLSTM architecture for flood prediction using self-synthesized dataset from CHIRPS, SRTM, HydroSheds, and Global Flood Database with spatio-temporal augmentation techniques	2026
Flood Detection Using Deep Learning UNet++, EfficientNet, TensorFlow <i>(In Progress)</i> • Training UNet++ with EfficientNet backbone for binary flood mask generation on Sen1Flood11 dataset	2026